

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

TAEHONG KWON

Student number: 28576810

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SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

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6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives. The tables of Chapter 4 form the record.

6.1 Answers to the research questions

The thesis proposed EndorseRank, PageRank with the edge weights of Adaptive Weighted PageRank (AWP) on a latest-allowance endorsement graph, and evaluated it together with AWP itself^[3], a coupled walk over both edge types and the raw degrees that these walks smooth. The evidence comes from GMX V2 wallets and from spenders on Arbitrum One between December 2025 and May 2026, with a registered replication on June to August 2026.

RQ1 asked whether EndorseRank and AWP rank the wallets of one sample differently. They do, and H1 is supported (Section 4.4). Much of the difference, and of the agreement, comes from EndorseRank’s tie blocks, so the claim that the two reflect different kinds of reputation rests mainly on the holdouts of RQ4.

RQ2 asked how closely each score agrees with allowance and transfer statistics in the same window. Each score follows the edges it walks on (Table 4.3). That supports H2 in a narrow form, because the proxies come from the same events as the graphs, and since EndorseRank’s coefficients are set by its ties, the thesis draws no conclusion from the difference between its allowance and transfer coefficients.

RQ3 asked what each score costs to compute. The cost follows the edge count, so EndorseRank, on the sparsest graph, is the cheapest score, and the coupled walks cost more than AWP (Table 4.1); H3 is supported. The pool stages could not test edge growth, because the second-tier extraction was not completed (Table 4.2).

RQ4 asked whether scores frozen at t_1 predict new approvals and new transfer senders on the spenders at t_2 , and whether any PageRank predicts them better than the raw degree it smooths. The first part holds, and the edge type decides which label a score anticipates (Table 4.10). The second does not: the raw in-approve degree predicts new approvals, and the raw transfer in-degree new senders, better than every walk in both windows (Tables 4.11 and 4.16). For these labels propagation adds no predictive value over the local count. The holdouts test the temporal predictive validity of each construct and leave its credit-risk validity untested.

RQ5 asked whether a coupled PageRank over both edge types improves on a walk over the transfers alone, and whether it ranks wallets better than either layer walked alone. Both answers are no, each under a rule fixed in advance. Under the rule of 14 September, C-PR fails both conditions of the primary part and the allowance condition of the secondary part (Section 4.3.1; Table 4.11). Its exploratory gain over the transfer walk on new approvals in the spring replicated in the registered June–August window, but a loss on new senders larger than the registered margin could not be ruled out, so the first answer is no as well; against AWP the loss on new senders was clear (Table 4.15). The seeded ablation matches the transfer walk: allowance edges contribute when the walk moves along them, and not as a seed.

Claims C1–C5 (Section 4.7) state these findings formally. Which score to deploy depends on whether the user’s question concerns authorization or flow, and on whether the user can run two scores or is tied to one.

6.2 Original contribution to knowledge

Section 1.9 named four contributions. In the terms of the literature reviewed in Chapter 2, this thesis establishes the following.

Before this work, graph reputation on public ledgers ranked wallets by their transfers ^{[3],[4]}, by how risk spreads through transaction networks ^[12], or by explicit ratings and stakes ^{[25],[26]}, and the reputation tooling deployed in DeFi scored the holder of an address from credentials ^[64]. None of it

used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power^[18], as a graph edge, and none evaluated on-chain reputation scores on one cohort against the checks of each construct and against the raw degrees they smooth, with confidence intervals, difference tests and a temporal holdout. This thesis supplies the allowance edge and that evaluation. PageRank and AWP’s weighting are taken from the literature^{[3],[15]}; EndorseRank changes only the edge, and the coupled operator changes the walk.

The new knowledge has three parts. What an on-chain reputation score measures depends on the edges it uses, and this can be verified on one cohort with quantified uncertainty. The allowance edge yields a score that is cheaper to update, differs from the transfer-based score, and ranks later endorsements of spenders in two windows, a label on which the scores built on transfers alone stay low; it carries little information about inflow or about traders. And two further steps that the literature suggested did not improve results on these data: propagation predicts the future growth of the graph no better than the raw degrees it smooths, and a weighted combination of the two edge types beats neither its better single layer nor, under the registered rule, the transfer-only walk. The construct-validity framing^{[22],[33]} that makes these claims testable is part of the contribution, as is the open pipeline whose published summaries let others check every table, since whether decentralized systems miscalculate credit scores is a matter of public interest^[2].

The scope is deliberately narrow. The thesis does not show that on-chain reputation can replace collateral^{[1],[9]} or that the results apply beyond Arbitrum One and the window studied, and it does not claim that EndorseRank resists Sybils: under uniform teleportation a closed farm of attacker addresses can inflate it cheaply (Section 5.6).

6.3 Future work

Each suggestion below corresponds to a limitation identified in Chapter 5.

- Complete the second-tier extraction. So far the expanded address pool has been used only to vary the number of nodes. With the approval and transfer logs of the supplemental addresses

extracted, we could measure how the algorithm’s running time grows with the number of edges, and the bounded claim for RQ3 would become a general scaling claim.

- Validate the results in other contexts. Running the same analysis on other rollups and on Layer-1 networks would help show whether each score’s agreement with its own construct and the holdout results extend beyond Arbitrum One. Because DeFi is heterogeneous ^{[1],[8]}, this is the main threat to external validity.
- Extend the time window and use a credit-linked holdout. A longer observation window would show whether the Kendall τ coefficients remain stable across different market conditions. A credit-linked holdout on the owners, feasible where lending protocols cover enough of the cohort, would show whether either graph carries credit-risk information, something the current holdouts do not show.
- Value weights. With $b = 1$ on raw base units, AWP’s value weight counts every approval or transfer of a few base units or more about once. A value of b set on price-normalized amounts would let size matter again, and the spring holdout suggests that it might: the allowance layer weighted by raw amounts ranked both spender labels better than EndorseRank. Such a choice should be fixed before it is scored.
- Adversarial evaluation. The Sybil cost model is analytical, backed by computations with the thesis’s own PageRank code on the observed EndorseRank graph (Section 5.6); no attack was carried out on chain. Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does ^[17], would remove the uniform inflow that closed farms exploit, and it should be evaluated under the same attacks. Empirical red-teaming with wash approvals, coordinated spender farms and strategic revocation would measure the realized cost-effectiveness of attacks under live fee markets and detection heuristics ^[11], and would test whether the Sybil-adjusted proxies remain informative under active adversaries.
- Coupling beyond a fixed transition-level mixture. Only a fixed λ with uniform teleportation on raw-amount layers was tried. Node-dependent or learned weights, per-layer damping,

mixtures of the two stationary vectors, and layers weighted as in EndorseRank and AWP remain untested. Any such design should fix its rules before its scores are computed, register a replication on new data, report construct-specific checks and time-split evidence against degree baselines, and keep the auditability that sets graph propagation apart from opaque learned scores ^{[2],[14]}.

- **Beating the degree.** On the growth of their own construct, every PageRank of this thesis is out-predicted by the raw degree of its own edge type at t_1 . What the holdout of this thesis leaves open for the field is whether any propagated score, on a label that is not itself a count of arrivals, adds out-of-window value over the local count.
- **Richer authorization semantics.** Where ERC-20 approvals are sparse, adding protocol-specific delegation logs to the edge definition may improve coverage, and the same reporting discipline would apply.
- **Composition with attestation tooling.** A Passport-style gate ^[64] answers whether an address belongs to a distinct person, and EndorseRank orders addresses by the authority others delegate to them. Filtering addresses through such a gate before ranking them is a plausible deployment, and it should be evaluated on labeled Sybil data, with the gate's own weaknesses ^{[65],[66]} in view.
- **Deployment studies.** Integrating EndorseRank with risk oracles and governance tooling would test whether its runtime profile can sustain production refresh cadences. Controlled experiments on fee tiers or monitoring cadence would let evidence replace the adoption hypotheses of Section 5.4.4. Collateral reduction remains a hypothesis and is not claimed by this research.

6.4 Closing reflection

In adversarial, pseudonymous markets on-chain reputation does not remove the need for collateral ^[9], but it can make behavioral structure readable, and which structure it reads depends on the edges

used. A transfer records economic flow and an allowance records delegated authority. PageRank will rank either graph, yet one ranking does not serve both constructs, and a walk over both graphs did no better on either than the right single graph. For a protocol or researcher who needs a fast, auditable signal of which spenders owners trust, the allowance edge is the one to use on data like those studied here. The approval count is its cheapest form, and EndorseRank is the propagated form for those who want the endorsers to count as well, with its exposure to cheap inflation by closed farms in mind (Section 5.6). For describing transfer hubs and anticipating their inflow, the transfer edge is the one to use. Whether trading profit can be predicted from these graphs alone is a question these data do not settle.